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MODULE LEADER: David Grey

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ASSIGNMENT: 1

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Full-Stack Development

Full-Stack Development is about creating both the end and back-end parts of a web application. A full-stack developer is someone who knows how everything works together from the buttons you click on a website to how that action changes the information in a database.

The Architecture of Full-Stack

Think of a web application like an iceberg. The part you see is what you interact with and the big part underneath is what makes it all work.

1. The Front-End

This is what you see and use when you visit a website.

- The main things it uses are HTML to make the structure CSS to make it look good and JavaScript to make it interactive.
- There are also ways of doing things like React, Vue.js and Angular.
- The goal is to make something that looks great loads quickly and works well on both computers and phones.

2. The Back-End

This is the part that makes the front-end work. You do not see it. It handles the stuff like security and data.

- The main things it uses are Node.js, Python, PHP, and Java or C #.
- There are also frameworks like Express.js for Node, Django for Python and Laravel for PHP.
- The goal is to keep users safe make sure the information they enter is correct handle file uploads and decide what information to show or update.

3. The Database Layer

This is where the website stores its information safely. When you sign up for something post a comment or buy something that information gets stored here.

- There are databases like MySQL and PostgreSQL that store information in tables.
- There are also -relational databases like MongoDB that store information in a more flexible way.

How Full-Stack Works in Practice

Let's say you fill out a form to sign up for something and click "Register":

1. Front-End: You type your name into the box. Click submit. JavaScript checks to make sure you entered something.

2. The Network Request: Your browser sends the information to a server.

3. Back-End: The server gets the request. Makes sure your password is safe. It checks the information. Gets ready to put it in the database.

4. Database: The server puts the information into the database. The database says "okay it's saved".

5. Full Circle: The server gets the message from the database. Sends a message back to your browser. The front-end shows a message that says "welcome".

- **Server** = the computer/environment where your application runs.
- **DevOps** = the tools, processes, and automation used to build, deploy, update, monitor, and maintain that application.

Popular Full-Stack Ecosystems

Developers like to use a group of technologies that work together called a "stack". One popular stack is MERN:

- MongoDB for the database
- Express.js for the server
- React for the end
- Node.js for the back-end

Using a stack like MERN means you can write all the code using JavaScript, from the database, to the front-end.